CS4084
Quantum Computing
Week 04 – Lecture 07-08

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Outline

1. Quick Review: What We Covered Last Lecture
2. Probabilistic States
3. Corelation Among the system : Concept of Independence
4. Tensor products of vectors
5. Tensor products of Standard Basis Vectors
6. Summary!

Quick Review: What We Covered Last Lecture

- **Unitary Evolution:** Quantum operations are reversible, norm-preserving transformations represented by 2×2 matrices for single qubits.
- **Composition of Operations:** Multiple gates applied in sequence correspond to matrix multiplication, where the algebraic order (right-to-left) must match the circuit order.
- **From Single to Multiple Systems:** Multiple systems are described by the combinations of their individual states.
- **Classical State Spaces:** For multiple classical systems, the combined state set is the **Cartesian Product** $(X \times Y)$ of the individual state sets.
- **Notation & Convention:** String notation (e.g., xy or 01) as a shorthand for the ordered pairs in a Cartesian product to represent joint states.

Key Message: Dirac Notation Unifies quantum states, unitary transformations, and circuit composition into a single mathematical framework.

Probabilistic States represented by Vectors

Probabilistic states of compound systems associate probabilities with the Cartesian product of the classical state sets of the individual systems.

Example

This is a probabilistic state of a pair of bits (X, Y):

$$\begin{aligned}
\Pr\big((X, Y) = (0, 0)\big) &= \tfrac{1}{2}\\
\Pr\big((X, Y) = (0, 1)\big) &= 0\\
\Pr\big((X, Y) = (1, 0)\big) &= 0\\
\Pr\big((X, Y) = (1, 1)\big) &= \tfrac{1}{2}
\end{aligned}$$

Continue..

$$\begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{pmatrix} \begin{array}{l} \leftarrow \text{probability associated with state 00} \\ \leftarrow \text{probability associated with state 01} \\ \leftarrow \text{probability associated with state 10} \\ \leftarrow \text{probability associated with state 11} \end{array}$$

Corelation Among the system

Definition

For a given probabilistic state of (X, Y), we say that X and Y are *independent* if

$$\Pr((X, Y) = (a, b)) = \Pr(X = a)\Pr(Y = b)$$

for all $a \in \Sigma$ and $b \in \Gamma$.

Suppose that a probabilistic state of (X, Y) is expressed as a vector:

$$|\pi\rangle = \sum_{(a,b)\in\Sigma\times\Gamma} p_{ab}|ab\rangle$$

The systems X and Y are independent if there exist probability vectors

$$|\phi\rangle = \sum_{a\in\Sigma} q_a|a\rangle \quad \text{and} \quad |\psi\rangle = \sum_{b\in\Gamma} r_b|b\rangle$$

such that $p_{ab} = q_a r_b$ for all $a \in \Sigma$ and $b \in \Gamma$.

Example...

Example

The probabilistic state of a pair of bits (X, Y) represented by the vector

$$|\pi\rangle = \frac{1}{6}|00\rangle + \frac{1}{12}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{4}|11\rangle$$

is one in which X and Y are independent. The required condition is true for these probability vectors:

$$|\phi\rangle = \frac{1}{4}|0\rangle + \frac{3}{4}|1\rangle \quad \text{and} \quad |\psi\rangle = \frac{2}{3}|0\rangle + \frac{1}{3}|1\rangle$$

Example...

Example

For the probabilistic state

$$\frac{1}{2}|00\rangle + \frac{1}{2}|11\rangle$$

of two bits (X, Y), we have that X and Y are not independent.

If they were, we would have numbers q_0, q_1, r_0, r_1 such that

$$\begin{aligned} q_0 r_0 &= \tfrac{1}{2} \\ q_0 r_1 &= 0 \\ q_1 r_0 &= 0 \\ q_1 r_1 &= \tfrac{1}{2} \end{aligned}$$

But if $q_0 r_1 = 0$, then either $q_0 = 0$ or $r_1 = 0$ (or both), contradicting either the first or last equality.

Mathematical Description of the Tensor Products of Vectors

Definition

The *tensor product* of two vectors

$$|\phi\rangle = \sum_{a\in\Sigma} \alpha_a |a\rangle \quad \text{and} \quad |\psi\rangle = \sum_{b\in\Gamma} \beta_b |b\rangle$$

is the vector

$$|\phi\rangle \otimes |\psi\rangle = \sum_{(a,b)\in\Sigma\times\Gamma} \alpha_a \beta_b |ab\rangle$$

Equivalently, the vector $|\pi\rangle = |\phi\rangle \otimes |\psi\rangle$ is defined by this condition:

$$\langle ab|\pi\rangle = \langle a|\phi\rangle\langle b|\psi\rangle \qquad (\text{for all } a \in \Sigma \text{ and } b \in \Gamma)$$

Example..

Definition

$$|\phi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle \quad \text{and} \quad |\psi\rangle = \sum_{b \in \Gamma} \beta_b |b\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = \sum_{(a,b) \in \Sigma \times \Gamma} \alpha_a \beta_b |ab\rangle$$

Example

$$|\phi\rangle = \frac{1}{4}|0\rangle + \frac{3}{4}|1\rangle \quad \text{and} \quad |\psi\rangle = \frac{2}{3}|0\rangle + \frac{1}{3}|1\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = \frac{1}{6}|00\rangle + \frac{1}{12}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{4}|11\rangle$$

Alternate Representation of Tensor Product!

Alternative notation for tensor products:

$$|\phi\rangle|\psi\rangle = |\phi\rangle \otimes |\psi\rangle$$

$$|\phi \otimes \psi\rangle = |\phi\rangle \otimes |\psi\rangle$$

Tensor Product of Column Vectors!

Following our convention for ordering the elements of Cartesian product sets, obtain this specification for the tensor product of two column vectors:

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} \otimes \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} = \begin{pmatrix} \alpha_1 \beta_1 \\ \vdots \\ \alpha_1 \beta_k \\ \alpha_2 \beta_1 \\ \vdots \\ \alpha_2 \beta_k \\ \vdots \\ \alpha_m \beta_1 \\ \vdots \\ \alpha_m \beta_k \end{pmatrix}$$

Tensor Product of Column Vectors!

Example

$$\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix} \otimes \begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\\ \beta_4\end{pmatrix} = \begin{pmatrix}\begin{pmatrix}\alpha_1\beta_1\\ \alpha_1\beta_2\\ \alpha_1\beta_3\\ \alpha_1\beta_4\end{pmatrix}\\ \begin{pmatrix}\alpha_2\beta_1\\ \alpha_2\beta_2\\ \alpha_2\beta_3\\ \alpha_2\beta_4\end{pmatrix}\\ \begin{pmatrix}\alpha_3\beta_1\\ \alpha_3\beta_2\\ \alpha_3\beta_3\\ \alpha_3\beta_4\end{pmatrix}\end{pmatrix}$$

Tensor products of Standard Basis Vectors

Observe the following expression for tensor products of standard basis vectors:

$$|a\rangle \otimes |b\rangle = |a\rangle|b\rangle = |ab\rangle$$

Alternatively, writing (a, b) as an ordered pair rather than a string, we could write

$$|a\rangle \otimes |b\rangle = |(a, b)\rangle$$

but it is more common to write

$$|a\rangle \otimes |b\rangle = |a, b\rangle$$

(It is a standard convention in mathematics to eliminate parentheses when they do not serve to add clarity or remove ambiguity.)

Tensor Products Properties

Important property of tensor products

The tensor product of two vectors is *bilinear.*

1. Linearity in the first argument:

$$\big(|\phi_1\rangle + |\phi_2\rangle\big) \otimes |\psi\rangle = |\phi_1\rangle \otimes |\psi\rangle + |\phi_2\rangle \otimes |\psi\rangle$$

$$\big(\alpha|\phi\rangle\big) \otimes |\psi\rangle = \alpha\big(|\phi\rangle \otimes |\psi\rangle\big)$$

2. Linearity in the second argument:

$$|\phi\rangle \otimes \big(|\psi_1\rangle + |\psi_2\rangle\big) = |\phi\rangle \otimes |\psi_1\rangle + |\phi\rangle \otimes |\psi_2\rangle$$

$$|\phi\rangle \otimes \big(\alpha|\psi\rangle\big) = \alpha\big(|\phi\rangle \otimes |\psi\rangle\big)$$

Notice that scalars "float freely" within tensor products:

$$\big(\alpha|\phi\rangle\big) \otimes |\psi\rangle = |\phi\rangle \otimes \big(\alpha|\psi\rangle\big) = \alpha\big(|\phi\rangle \otimes |\psi\rangle\big) = \alpha|\phi\rangle \otimes |\psi\rangle$$

Tensor products of Three or More Vectors

Tensor products generalize to three or more systems.

If $|\phi_1\rangle, \ldots, |\phi_n\rangle$ are vectors, then the tensor product

$$|\psi\rangle = |\phi_1\rangle \otimes \cdots \otimes |\phi_n\rangle$$

is defined by the equation

$$\langle a_1 \cdots a_n|\psi\rangle = \langle a_1|\phi_1\rangle \cdots \langle a_n|\phi_n\rangle$$

Equivalently, the tensor product of three or more vectors can be defined recursively:

$$|\phi_1\rangle \otimes \cdots \otimes |\phi_n\rangle = \big(|\phi_1\rangle \otimes \cdots \otimes |\phi_{n-1}\rangle\big) \otimes |\phi_n\rangle$$

Summary: Key Takeaways

- **Unitary Evolution:** Quantum operations are reversible, norm-preserving transformations represented by 2×2 matrices for single qubits.
- **Composition of Operations:** Multiple gates applied in sequence correspond to matrix multiplication, where the algebraic order (right-to-left) must match the circuit order.
- **From Single to Multiple Systems:** Just as single systems have state sets, multiple systems are described by the combinations of their individual states.
- **Classical State Spaces:** For multiple classical systems, the combined state set is the **Cartesian Product** ($X \times Y$) of the individual state sets.
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Next Lecture: Moving from Classical Cartesian Products to Quantum **Tensor Products**, Entanglement, and the Postulates of Multiple